EXPERIMENTS PROCEDURE

Full Explanation on *TITRATION*

Titration is a process whereby one solution(acid) is added gradually to another solution(base) until end point is reached.

End point is the point where the acid neutralizes the base or viceversa.

During titration, a phenolphthalein indicator is used.
Phenolphthalein is pink in colour.
When added to a base,the mixture

turns pink but when phenolphthalein is added to an acid the mixture turns red.

During the titration experiment, one solution must have its concentration known. A solution with known concentration is called a *Standard solution*
Usually, (but not always), Sodium Hydroxide (NaOH) has its concentration known.
And Hydrochloric acid (HCl) has unknown concentration.

After titration we need to calculate

the molarity or concentration of the unknown thats how much has been used to neutralize the standard solution.

Now the whole experiment.
Suppose Sodium hydroxide (NaOH) is our standard solution and Hydrochloric acid is our unknown concentration.

i) Firstly, we add eg 20ml of NaOH (base) into a conical flask. Record the volume (eg 20ml)

ii). Using a dropper, we add 2 or 3

drops of Phenolphthalein indicator into the conical flask with NaOH.
The mixture will be pink in colour.

iii). Then we measure 20ml of HCl and we are told to add to the mixture of NaOH and Phenolphthalein indicator.
The adding should be done slowly by slowly or gradually while shaking the contents.
Stop adding the acid until the end point is reached thus the mixture has changed from pink into colourless.
This is the point where the acid has neutralized the base.

iv) Record the volume used to neutralize the base (e.g. if it was 20ml and mwatsala 13ml then 7ml has bn used in the reaction)

v) Calculate the concentration of unknown using the formula $C1V1=C2V2$, where C=concentration or molarity and V =Volume.

Note

During titration you maybe asked to write the balanced equation for the reaction so it is

$HCl + NaOH \text{ ----->} NaCl + H_2O$

This one is already balanced.

Some questions also follow after the experiment like

Source of error and how to reduce.

Error source could overshooting.

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FOOD TESTS

Food substances required: solution

of glucose, sucrose and starch;
ground-up.
suspension in water of
pea,bean,carrot, and grape
;milk,castor -oilbeans, and
cookingoil;
pieces of potato, bread ,and boiled
white of egg
(Note. Sometimes you are provided with the real food stuff koma Pena you are given the food solution but the procedure is the same)

1- *Test for Starch*

-Method

place about 3cm of
starch solution in a
container
add 2-3drops of iodine
solution
observe colour change

-Results
blue-black
color
means(presence of
starch
)
browncolor means (absence of
starch
)

2- *Test for Protein*

- Method

place about $3cm^3$ of the food solution in a test tube

carefully add 4-5drops of biuret solution.

-Results

purplecolor means(presence of protein
)

blue colour means (absence of protein
)

ii- *Test for Protein*

- Method

placea bout $3cm^3$ of food solution in a test tube

add about $3cm^3$ of sodium hydroxidesoln.and mix well

-Add 1or 2drops of coppersulphate solution mix well and observe any colour change

NB:if the colour does not change immediately let the solution stand for 15minutes and then observe again

-Results

violet color (presence of protein)

3- *Test for Lipids*

-Method

place adrop of oil/food substance on apiece of paper (filter paper)

examine the paper while holding up against light

-Results

translucent
or grease
spot
means (presence of
lipids
)
not
transparent
means (absence of
lipids)

ii- *Test for Lipids*

2drops of cooking oil are
thoroughly shaken with
about 5cm³ of ethanol in a

dry test tube until the fat

dissolves.

 the alcoholic solution is

poured into a test tube containing

afew water

(NB:this is an ethanol/emulsion

test

)

-Results

 cloudy

white

emulsion

 means (presence of

fats/lipids

)

if the liquid

stay clear

means (absence of

fats

)

4- *Test for Reducing

sugars*

(monosaccharides .eg

glucose

and

maltose)

-Method

place about 3cm³ of the

food solution in a test tube

add about 2drops of

benedict solution

heat gently for few

minutes

(NB:the mouth of the test-tube

must be pointed away from

people as the solution tend to bump

out of the tube)

-Results

Ayellow or

orange or

brickred

color

means(presence of

reducing

sugar

)

blue colour meeans (absence of

reducing

sugar

)

ii- *Test for Reducing Sugars*

- Method

place aboutb3cm³ of the

food solution in a test tube

Add about 2cm³ of

Fehling’s A solution followed

by the same Fehling’sB
solution.
 heat gently for few
minutes

-Results
 Ayellow or
orange or
brickred
color
means(presence of
reducing
sugar),,,

EFFECTS OF PRACTICE ON HITTING THE TARGET

Procedure

1-Put a mark on the wall

2-measure a distance of 10 m from the mark ànd draw a line

3- identify a person and give him 10 small objects such as darts or stones

4-let the person stand on the line and throw one object at a time towards the mark

5-count the number of objects that hit the mark and record it

6- repeat the same activity with the

same person for three more trials .
7-compare the number of objects that hit the mark in all trials .

Expcted results
More objects will hit the mark in the last trial than the first trial

Conclusion
Practice increases chances of hitting the target

EFFECTS OF TIME OF THE DAY ON MEMORISING LIST OF WORDS

Procedure

1- write a list of different words of same length and level of difficult on three chart papers

2- Identify the person who will recite the words on the paper.

3-flash the first paper to a person for a minute in the morning and hide it

4-Ask the person to recite the words of the hidden paper.

5-Record the number of words that are recited correctly by the person

6- Repeat the same activity with the same person using the second and third paper at noon and in the afternoon respectively

7-Compare the results obtained in three trials .

Expected results

More words will be corectly recited in the morning than at noon and in the afternoon

Conclusion

People memorise more words in the morning than at noon and in the afternoon

*EXPERIMENT TO SHOW THAT A

GERMINATING BEAN SEED CONTAINS ENZYMES THAT DIGEST STATCH*

Procedure

1. crush the germinating bean seed in a mortar
2. add water to obtain(enzyme) extract
3.put starch soln. in test tube A and B
4.in a test tube A add extract ,but leave test tube B in intact
5.leave both test tube to stand for some time
6.later add drops of iodine soln. to

both test tubes and observe color change

Results

In test tube A-there will be brown color. This show that there’s no starch present

In test tube B –you will observe blue black color. This change indicate presence of
starch

conclusion

Germinating bean seeds contain an enzyme which digested starch in test

All de best

@Mr President

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